

# Assessment 1 Report – Do Thuy Linh Le 48625610

## 1. System overview

### Pipeline

The Retrieval-Augmented Generation (RAG) pipeline consists of data loading, chunking, embedding, storage, retrieval, and generation.

A Document class with attributes text and metadata was implemented. Raw text files were parsed into dictionaries containing metadata headers and body text, then loaded as a list of documents. Before chunking, text was cleaned by removing unnecessary whitespace and newline characters.

Documents were split using a chunk size of 180 tokens with a 40-token overlap. The resulting chunks were embedded using both BM25 and a SentenceTransformer model. Embeddings were then stored in a ChromaDB vector database to support efficient retrieval.

### Retrieval Methods Implemented

**BM25:** The BM25 retriever tokenises the query in the same way as the corpus. Scores are then calculated using the BM25 index based on term frequency and inverse document frequency. These scores are sorted and the most relevant chunks are returned.

**Dense Retrieval:** Dense retrieval computes the cosine similarity between the query embedding and the embeddings of all document chunks. The closest embeddings are identified, sorted by similarity score, and returned as a ranked list of relevant Document objects.

**ChromaDB Vector Database Retrieval:** In this method, the query embedding is sent directly to ChromaDB. The vector database compares it against stored embeddings and returns the top-k most similar chunks.

**Hybrid RRF:** The hybrid retriever combines BM25 and ChromaDB rankings using Reciprocal Rank Fusion (RRF). This method merges the rankings from both retrieval systems and selects the most relevant results based on their combined scores.

## 2. Chunking and indexing choices

A chunk size of 180 tokens with a 40-token overlap was selected. Smaller chunks increase retrieval precision, which is useful for search and question-answering tasks that require locating specific information.

The 22% overlap preserves context between neighbouring chunks, reducing the risk of splitting important information across boundaries.

Since many corpus files are already close to this length, most documents remain intact after chunking and little contextual information is lost.

### 3. Retrieval methods compared

Each query from public\_eval.json was tested with BM25, dense retrieval, ChromaDB retrieval, and hybrid RRF. The top three results from each retriever were stored in retrieval\_results.json.

Results were compared with the gold documents provided in the dataset. Across most queries, at least one gold document appeared in the top three results for all methods. Frequently, the top two gold documents were retrieved consistently.

Dense retrieval and ChromaDB retrieval performed similarly and generally outperformed BM25. These methods often retrieved more gold documents and ranked them higher. Their advantage comes from semantic embeddings, which capture conceptual similarity rather than exact word matches.

BM25 performed slightly worse due to its reliance on lexical matching. It may miss relevant documents when synonyms or different wording are used.

Hybrid RRF showed lower consistency than expected. Although it combines sparse and dense retrieval, BM25's weaker rankings likely reduced the effectiveness of the fused results. Hybrid retrieval often included gold documents within its top three results but ranked them lower than other methods.

### 4. Evaluation results

Evaluation metrics were calculated using Precision@k, Recall@k, Hit@k, and Mean Reciprocal Rank (MRR). For all metrics, k was set to 3.

|   | method     | precision@k | recall@k   | hit@k | mrr        |
|---|------------|-------------|------------|-------|------------|
| 0 | bm25       | 0.61111111  | 0.84722222 | 1     | 0.91666667 |
| 1 | dense      | 0.75        | 1.02777778 | 1     | 1          |
| 2 | chroma     | 0.75        | 1.02777778 | 1     | 1          |
| 3 | hybrid_rrf | 0.55555556  | 0.77777778 | 1     | 0.95833333 |

Table 1: Evaluation results of each retrieval method

Dense and ChromaDB retrieval achieved identical scores and the highest precision and recall. BM25 scored lower across most metrics, while hybrid retrieval produced the lowest precision and recall but a slightly higher MRR than BM25.

## 5. Error analysis

Although retrieval methods behaved similarly for many queries, some errors occurred.

Query 1: Why is hybrid retrieval useful on this corpus?

**Gold documents:** doc05, doc28, doc23

|            | First document | Second document | Third document |
|------------|----------------|-----------------|----------------|
| BM25       | doc13          | doc05           | doc28          |
| Dense      | doc05          | doc28           | doc19          |
| ChromaDB   | doc05          | doc28           | doc19          |
| Hybrid RRF | doc05          | doc28           | doc13          |

Table 2: Retrieval results on Query 1

Irrelevant documents such as doc13 and doc19 appear in the results, while doc23 is missing entirely. BM25 ranked doc13 first, likely due to lexical matching. Because BM25 relies on exact word overlap, it may retrieve documents that share keywords but lack semantic relevance.

Dense retrieval and ChromaDB use vector embeddings that capture conceptual similarity, allowing them to match queries with semantically related documents.

Query 2: How should obsolete documents be handled in retrieval?

**Gold documents:** doc27, doc20

|  | First document | Second document | Third document |
|--|----------------|-----------------|----------------|
|--|----------------|-----------------|----------------|

|            |       |       |       |
|------------|-------|-------|-------|
| BM25       | doc13 | doc05 | doc28 |
| Dense      | doc05 | doc28 | doc19 |
| ChromaDB   | doc05 | doc28 | doc19 |
| Hybrid RRF | doc05 | doc28 | doc13 |

Table 2: Retrieval results on Query 1

None of the methods retrieved the correct documents. Instead, all returned similar incorrect results such as doc13, doc05, doc28, and doc19.

When all retrieval methods fail in the same way, the issue may originate earlier in the pipeline. Possible causes include chunking errors that remove key context, incorrect metadata, or problems during data loading and indexing.

## 6. Groundedness / hallucination notes

### Example 1: Grounded and relevant answer

**Query:** How many days of paid sick leave do employees receive each year?

**Retrieved context:** Full-time employees receive 10 days of paid sick leave per year. Unused sick leave does not roll over into the next year.

**Answer:** Employees receive 10 days of paid sick leave each year.

When these parameters were called with the `heuristic_faithfulness` function, the following was printed:

```
Faithfulness : {'support_ratio': 0.8888888888888888, 'flag': 'supported'}
```

This indicates that the generated answer was well supported by the retrieved context. The support ratio represents the proportion of answer tokens that also appear within the provided context.

### Example 2: Partially relevant but hallucinated answer

**Query:** What is the remote work policy?

**Retrieved context:** Employees may work remotely up to two days per week with manager approval. Remote work is not guaranteed for all roles.

**Answer:** Employees may work remotely five days per week and receive a home office allowance.

When the `heuristic_faithfulness` function was called with these inputs, the following was printed:

```
Faithfulness : {'support_ratio': 0.5833333333333334, 'flag': 'weak'}
```

This issue may arise from both retrieval and generation factors. If the correct information is not retrieved, the model may attempt to infer a plausible answer. Additionally, the prompt currently does not include strict instructions to avoid hallucination. For example, a prompt rule such as “If the answer is not in the context, respond that the information is unavailable” could reduce unsupported responses.

## 7. Conclusion

This project implemented and evaluated a local RAG system for OrbitOps using four retrieval methods: BM25, dense retrieval, ChromaDB vector retrieval, and hybrid RRF.

Dense and ChromaDB retrieval performed best across evaluation metrics, likely due to their semantic understanding of queries and documents. BM25 showed weaker performance because it relies on exact word matching. Hybrid retrieval did not significantly improve results, suggesting that weaker BM25 rankings affected the fused output.

Overall, the results highlight the effectiveness of semantic retrieval methods for RAG systems, particularly when handling natural language queries and conceptually related documents.